Banking Credit Risk Calculator

Solution Walkthrough

Credit Risk & Banking Intelligence Platform

Banking Credit Risk Calculator — Internal Solution Overview

July 2026

Outline

Our Banks — Portfolio Summary (FY2026)

Nine banks, grounded in a real per-loan ledger — every figure below is computed directly from `bank.db` (balance sheet + loan book), not estimated.

Bank	Country	Total Assets (Rs. Cr)	Total Liabilities (Rs. Cr)	Loans & Advances (Rs. Cr)	Customers	Loans	Defaults
HDFC Bank	India	191.3	191.3	130.5	2,213	2,045	215
ICICI Bank	India	202.3	202.3	138.0	2,275	2,132	228
JPMorgan Chase N.A.	United States	173.6	173.6	118.4	2,277	2,112	215
Barclays Bank PLC	United Kingdom	150.7	150.7	102.8	2,419	2,215	216
DBS Bank Ltd	Singapore	201.9	201.9	137.7	2,341	2,169	211
Emirates NBD	UAE	190.4	190.4	129.8	2,339	2,169	216
Bank of Baroda	India	212.5	212.5	144.9	2,108	1,961	223
Commonwealth Bank	Australia	184.8	184.8	126.0	2,203	2,035	212
Punjab National Bank	India	206.8	206.8	141.0	2,448	2,317	216
Total (9 banks)	—	**1,714.5**	**1,714.5**	**1,169.2**	**20,623**	**19,155**	**1,952**

How these numbers are derived

Total Assets/Liabilities from `bank_balance_sheet` (FY2026, as-on 2026-03-31) — the two always tie out exactly, as required by double-entry accounting. Loans & Advances is the `advances_net` balance-sheet line (the active loan book). Customers and Loans are row counts from `customers` and `loans`. Defaults = loans classified `Written-Off` — the platform's realised-default definition.

The platform scores every borrower under one of four Basel exposure classes — each routed to its own PD model. Figures below are aggregated across all 9 banks.

Exposure Type	Customers	Loans	Defaults	Default Rate
CORPORATE	5,070	5,070	488	9.63%
SME	4,655	4,655	488	10.48%
RETAIL_MORTGAGES	4,950	4,950	488	9.86%
RETAIL_OTHER	4,479	4,480	488	10.89%
Total	**19,154**	**19,155**	**1,952**	**10.19%**

How these numbers are derived

Customers = distinct borrowers holding at least one loan in that exposure class (`loans.exposure_class`). Loans = row count in `loans` for that class. Defaults = loans classified `Written-Off` — the same realised-default definition used on the previous slide.

Model Training Attributes — Grouped by the Five C's

All 4 segment PD models (CORPORATE, SME, RETAIL_MORTGAGES, RETAIL_OTHER) are trained on the same 36-attribute schema (`ml_models/pd_model_metadata_*.json`).

5C	Feature	Description
Character	age	Borrower's age (years)
	employment_type_enc	Employment type (encoded)
	years_employed	Years in current job
	city_tier_enc	City tier (encoded)
	education_enc	Education level (encoded)
	cibil_score	Credit bureau score (300–900)
	months_as_customer	Relationship tenure with bank
	num_late_payments_past_12m	Late payments, past 12 months
Capacity	de_ratio	Debt-to-equity ratio
	interest_coverage	Interest coverage ratio
	annual_income	Declared annual income
	foir	Fixed obligation-to-income ratio
	num_dependents	Number of dependents
	loan_purpose_enc	Loan purpose (encoded)
	existing_loans_count	Count of existing loans
Capital	profitability	Net profit margin (%)
	liquidity_ratio	Current ratio (liquidity)
	residence_type_enc	Residence type (owned/rented)
	num_existing_products	Existing product holdings
Collat.	ltv_trend_pct	LTV trend since origination (%)

5C	Feature	Description
Conditions	gdp_growth_pct	GDP growth rate (%)
	inflation_cpi_pct	CPI inflation rate (%)
	policy_rate_pct	Central bank policy rate (%)
	unemployment_pct	Unemployment rate (%)
	delta_de_ratio	Change in leverage since origination
	delta_cibil	Change in credit score
	months_since_origination	Loan seasoning (months)
	delta_gdp_pct	YoY change in GDP growth
	delta_cpi_pct	YoY change in inflation
	delta_policy_rate_pct	YoY change in policy rate
	delta_unemployment_pct	YoY change in unemployment
	macro_regime_score	Composite macro distress score (0–100)
	ecs_bounce_count	EMI/ECS bounce count
	other_lender_emi_ratio	EMI ratio owed to other lenders
	income_disruption_flag	Income disruption flag (0/1)
	sector_stress_index	Sector-level stress index

Why Conditions dominates

Macro, trend, and behavioural-bureau signals change every observation period — they carry most of the schema's breadth even though Character/Capacity drive more of the model's actual feature importance.

Why XGBoost, and How We Deploy It

Why XGBoost (not a simpler model)

- Handles nonlinear interactions across mixed feature types (ratios, encoded categoricals, macro indicators) without manual feature engineering
- Tree regularisation (`min_child_weight=10`, `gamma=1.0`, `reg_alpha=1.0`, `reg_lambda=2.0`) controls overfitting on a minority-class (default) problem
- `scale_pos_weight=45` explicitly compensates for class imbalance
- Fast, exact SHAP (TreeExplainer) attribution — every prediction is explainable, not a black box
- Industry-standard for tabular credit data (Chen & Guestrin 2016; consistently top-performer in large benchmark studies)
- **Not just assumed** — the pipeline can swap in a logistic-regression baseline; XGBoost is kept because it beats that baseline with statistically significant lift (next slide)

Deployment plan — one model per exposure class

- 4 independent XGBoost models: CORPORATE, SME, RETAIL_MORTGAGES, RETAIL_OTHER — trained, evaluated, promoted separately
- Promoting one segment's model never touches another's active slot — independent rollback per segment
- Pipeline: ingest → validate → 80/20 split → train → evaluate (single-split + 5-fold CV) → statistical soundness checks → generate charts → promote if criteria met → archive prior model

Example output (illustrative)

SME applicant → PD = 6.8% → Grade **BB** (Speculative) → REFER → LGD 40% → EL 2.7% of EAD → indicative rate ≈ 11.2% → SHAP top factors: FOIR (+), CIBIL score (−), years employed (−)

How We Know the Model Is Good Enough

Segment	AUC-ROC	5-Fold CV AUC	Accuracy	Recall	EPV	Baseline AUC	Lift (95% CI)	Significant?
CORPORATE	0.754	0.765 ± 0.013	60.9%	73.6%	11.0	0.622	+0.132 [0.051, 0.214]	Yes (p=0.002)
SME	0.764	0.746 ± 0.019	63.7%	79.4%	10.9	0.654	+0.110 [0.046, 0.180]	Yes (p<0.001)
RETAIL_MORTGAGES	0.758	0.746 ± 0.018	63.0%	76.5%	10.7	0.637	+0.121 [0.061, 0.181]	Yes (p<0.001)
RETAIL_OTHER	0.753	0.767 ± 0.022	65.4%	78.8%	10.8	0.641	+0.112 [0.036, 0.186]	Yes (p=0.006)

- **AUC-ROC / 5-fold CV AUC** — discriminative power on one split vs. averaged across 5 splits (catches a model that got lucky/unlucky on one split)
- **Recall** — of actual defaulters, how many are caught — business-critical, since missed defaulters are direct losses
- **EPV (Events Per Variable)** — defaulting examples ÷ feature count; ≥10 acceptable, ≥20 comfortable (Peduzzi et al.) — guards against more model freedom than the data supports
- **Baseline lift test** — bootstrap significance test (1,000 resamples) vs. a trivial CIBIL-only logistic regression; a significant positive lift proves the other 35 features add real signal, not overfitting capacity

New Customer: What We Ask For

Single origination channel

One intake form (`borrower-info.html`) serves every customer type. Data is captured **once** and every department downstream (Operations, Regulatory, RM, Financials) reads from it — no re-entry, no reconciliation.

Common fields (all customers)

- Borrower ID / Name, Exposure Amount, Country
- KYC status, sanctions screening result
- Proposed EMI, existing monthly obligations, sector
- **Exposure Class** — routes to the matching PD model
- Loan Type (auto-sets maturity), Collateral Type/Value, Seniority, LTV

Segment-specific fields

- **Retail** (Mortgages/Other): age, employment type, years employed, annual income, FOIR, dependents, city tier, education, residence type, loan purpose, CIBIL score, previous-default flag, months as customer, late payments (12m), existing loans/products, rural flag
- **Corporate/SME**: the 4 financial ratios that feed the model directly — Debt-to-Equity, Interest Coverage, Profitability Margin, Liquidity Ratio

Shortcut for repeat borrowers

Existing-customer lookup by Customer ID autofills the entire form from bank records — no re-typing.

From Data to Decision: Analysis & Presentation

PD → Rating → LGD → RWA → Exp. Loss → Pricing → Five C's → Policy → Machine Reco.

How it's explained, not just scored

- SHAP attribution — top factors pushing PD up/down, in plain language
- Peer comparison — segment-scoped benchmark against approved borrowers in the same exposure class
- Counterfactual recourse — "what would need to change to qualify" for borderline/declined applicants

How it's presented — two views, one finding

- **Underwriter report** (internal, 6 sections): Risk Assessment, Five C's, Peer Comparison, Improvement Pathways, AIRB Regulatory Detail, Credit Decision
- **Applicant letter** (adverse-action style) — decision and reason, not the technical internals

Output object

The **Machine Recommendation (M)** bundles the risk estimate, policy result, confidence/sensitivity, and required-control-level routing into one hash-stamped record — this is what the RM reviews next.

Role of the Relationship Manager

Where the RM sits

The Human (H) node between the **Machine Recommendation (M)** and the **Org Decision (O)** — the RM does not replace the model, and the model does not replace the RM.

What the RM actually does

- Reviews the Machine Recommendation — SHAP factors, peer comparison, counterfactual recourse, policy flags
- **Two actions only** — Accept or Reject, each requiring a $\geq$20-character rationale; both finalise immediately, no partial states
- Whether the RM can decide *alone* depends on the DoA control level already computed (`RM_SINGLE` / `FOUR_EYES` / `COMMITTEE` / `AUTO_DECLINE`) — the platform tells the RM up front whether this case is even theirs to close
- After a decision, the RM can trigger PDF report generation; every action is hash-chained into the case's audit trail

What the RM does *not* do

- Re-enter borrower data
- Override the model's PD
- Approve outside their delegated authority

These are **structurally prevented**, not just discouraged by policy.